Global Identification Impossibility in Feedback-Stabilized Economic Systems

Extended Version

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Abstract

We establish a *global identification impossibility theorem* for economic systems governed by fixed-point dynamics under feedback control. Even with complete knowledge of the induced closed-loop operator and access to an infinite sample, the underlying structural mapping and policy rule are not uniquely recoverable—a result we term *structural non-identification under feedback*. The impossibility is formalised using operator theory on Banach function spaces, and constructive functional arguments yield explicit infinite-dimensional families of observationally identical models. We extend the core result in four directions: *(i)* a sharp characterisation of the equivalence class of observationally identical structural pairs via a Fréchet-derivative argument; *(ii)* statistical implications showing that standard nonparametric estimators consistently recover only the closed-loop operator, together with minimax lower bounds; *(iii)* impossibility results for machine-learning-based structural identification and deterministic algorithmic recovery; and *(iv)* a discussion of the minimal exogenous variation—captured through the do-calculus of Pearl [12]—required to restore point identification. The results formalise and substantially generalise the Lucas [8] Critique under a functional-analytic lens, connecting it to the partial identification literature initiated by Manski [9].

The paper ends with "The End"

Keywords: identification, feedback control, fixed-point dynamics, structural econometrics, Lucas Critique, operator theory, partial identification, causal inference, machine learning.

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1 Introduction

A central challenge in structural econometrics is the recovery of primitive model objects—structural functions and policy rules—from observed equilibrium data. This problem is substantially harder when the system operates under a *feedback control loop*: the policy instrument is itself determined as a measurable function of the current state. Under such feedback the data-generating process is a composite object that conflates the structural dynamics and the policy rule in a way that, as we prove, cannot be disentangled by observation alone.

To fix ideas, consider a discrete-time dynamic economic system:

$$P_{t+1} = f(P_t,\, Z_t), \qquad Z_t = \pi(P_t), \tag{1}$$

where $P_t \in \mathcal{P} \subset \mathbb{R}^d$ is the state vector (prices, quantities, output), $Z_t \in \mathcal{Z} \subset \mathbb{R}^k$ is the policy instrument, $f\colon \mathcal{P}\times\mathcal{Z} \to \mathcal{P}$ is the *structural transition operator*, and $\pi\colon \mathcal{P} \to \mathcal{Z}$ is the *feedback policy rule*. Substituting the policy into the structural equation yields the *closed-loop map*:

$$\tilde{f}(P) \;=\; f\big(P,\, \pi(P)\big). \tag{2}$$

This is the only object visible to an econometrician observing the joint trajectory $\{(P_t, P_{t+1})\}_{t\geq 0}$. The identification question is: *does knowledge of $\tilde{f}$ uniquely pin down the pair (f, π)?*

The answer—our main result—is emphatically negative. The preimage of $\tilde{f}$ under the composition operator is *infinite-dimensional*, meaning that infinitely many structural pairs (f, π) generate identical observational data. This failure is not a matter of finite-sample limitations: it persists even with infinite data and perfect knowledge of $\tilde{f}$. It is a fundamental logical impossibility.

1.1 Related Literature

The identification of structural models is a central theme in econometrics. Rothenberg [14] established foundational *local* identification conditions in parametric settings. Manski [9, 10] developed the framework of partial identification, providing sharp bounds when point identification fails. Chesher [3] studied identification in nonseparable models, and Tamer [17] analysed incomplete models. Our paper contributes a *global, nonparametric* impossibility result driven by the feedback structure.

Lucas [8] argued that reduced-form relationships are not invariant to policy regime changes. Our result formalises this: $\tilde{f}$ is the stable object across regimes whenever π is fixed, but recovering f from $\tilde{f}$ without knowing π is impossible.

In control theory, closed-loop identification has been studied by Van den Hof and Schrama [18] and Ljung [7]. Our paper imports these insights into econometrics with attention to function-space structure and statistical implications. In machine learning, Peters et al. [13] and Schölkopf et al. [15] emphasise the limits of purely observational learning for structural recovery; our theorem provides a formal impossibility result in the economic feedback setting.

2 Function Space Setup

2.1 State and Instrument Spaces

Assumption 2.1 (Compactness)**.** $\mathcal{P} \subset \mathbb{R}^d$ and $\mathcal{Z} \subset \mathbb{R}^k$ are compact, convex sets with non-empty interiors.

Endow $C^1(\mathcal{P}\times\mathcal{Z}, \mathcal{P})$ and $C^1(\mathcal{P}, \mathcal{Z})$ with their standard C^1 topologies (uniform convergence of functions and all first partial derivatives on compacts). Define the *structural space* and *policy space*:

$$\mathcal{F} \;=\; C^1(\mathcal{P}\times\mathcal{Z},\, \mathcal{P}), \qquad \Pi \;=\; C^1(\mathcal{P},\, \mathcal{Z}). \tag{3}$$

Both are Banach spaces under their respective sup-norms; we write $\|\cdot\|_{\mathcal{F}}$ and $\|\cdot\|_{\Pi}$ for these norms.

2.2 The Composition Operator

Definition 2.2 (Composition operator)**.** The *composition operator* $\mathcal{C}\colon \mathcal{F}\times\Pi \to C^1(\mathcal{P}, \mathcal{P})$ is defined by

$$\mathcal{C}(f, \pi)(P) \;=\; f\big(P,\, \pi(P)\big).$$

Lemma 2.3 (Continuity of $\mathcal{C}$)**.** *$\mathcal{C}$ is jointly continuous in (f, π) under the C^1 topology on $\mathcal{F} \times \Pi$ and the C^1 topology on $C^1(\mathcal{P}, \mathcal{P})$.*

Proof. Let $(f_n, \pi_n) \to (f, \pi)$ in $C^1 \times C^1$. Then

$$|\mathcal{C}(f_n, \pi_n)(P) - \mathcal{C}(f, \pi)(P)| \leq |f_n(P, \pi_n(P)) - f(P, \pi_n(P))| + |f(P, \pi_n(P)) - f(P, \pi(P))|.$$

The first term tends to zero by uniform convergence of f_n. The second tends to zero by uniform continuity of f on the compact set $\mathcal{P} \times \mathcal{Z}$ and uniform convergence of π_n. Derivative estimates follow by the chain rule applied to each partial. □

Lemma 2.4 (Fréchet differentiability)**.** *$\mathcal{C}$ is Fréchet differentiable at every $(f, \pi) \in \mathcal{F} \times \Pi$, with derivative*

$$D\mathcal{C}(f, \pi)[\delta f, \delta\pi](P) \;=\; \delta f\big(P, \pi(P)\big) \;+\; \partial_Z f\big(P, \pi(P)\big) \cdot \delta\pi(P).$$

Proof. A Taylor expansion gives $\mathcal{C}(f+\delta f, \pi+\delta\pi)(P) = f(P, \pi(P)) + \delta f(P, \pi(P)) + \partial_Z f(P, \pi(P))\delta\pi(P) + O(\|\delta f\|^2 + \|\delta\pi\|^2)$, confirming Fréchet differentiability with the stated derivative. □

2.3 Observational Equivalence

Definition 2.5 (Identification fiber)**.** For any $\tilde{f} \in C^1(\mathcal{P}, \mathcal{P})$, the *identification fiber* (equivalence class) of $\tilde{f}$ is

$$E(\tilde{f}) \;=\; \big\{(f, \pi) \in \mathcal{F} \times \Pi : f(P, \pi(P)) = \tilde{f}(P)\ \forall P \in \mathcal{P}\big\}. \tag{4}$$

Our main results characterise the geometry and dimensionality of $E(\tilde{f})$. As we show, $E(\tilde{f})$ is an *infinite-dimensional affine submanifold* of $\mathcal{F} \times \Pi$, and no regularity conditions short of direct observation of counterfactual system responses can collapse it to a singleton.

3 Main Identification Results

3.1 Single-Equation Non-Identification

Theorem 3.1 (Non-identification given π)**.** *For any $\tilde{f} \in C^1(\mathcal{P}, \mathcal{P})$ and any $\pi \in \Pi$, the fiber*

$$F(\tilde{f}, \pi) \;=\; \{f \in \mathcal{F} : f(P, \pi(P)) = \tilde{f}(P)\ \forall P \in \mathcal{P}\}$$

is infinite-dimensional. Specifically, it contains a linear subspace isomorphic to $C^1(\mathcal{P} \times \mathcal{Z}, \mathbb{R})$.

Proof. Fix $\pi \in \Pi$ and $\tilde{f} \in C^1(\mathcal{P}, \mathcal{P})$. For any $\psi \in C^1(\mathcal{P} \times \mathcal{Z}, \mathbb{R})$, define

$$f_\psi(P, Z) \;=\; \tilde{f}(P) \;+\; \psi(P, Z) \cdot |Z - \pi(P)|^2. \tag{5}$$

Since $|Z - \pi(P)|^2$ vanishes on the graph $\mathrm{graph}(\pi) = \{(P, \pi(P)) : P \in \mathcal{P}\}$, we have

$$f_\psi(P, \pi(P)) \;=\; \tilde{f}(P) + \psi(P, \pi(P)) \cdot 0 \;=\; \tilde{f}(P) \quad \forall P \in \mathcal{P}.$$

Furthermore $f_\psi \in C^1$ since $\tilde{f}$, ψ, and the squared norm are all C^1. The map $\psi \mapsto f_\psi$ is linear and injective: if $f_{\psi_1} = f_{\psi_2}$, then $(\psi_1 - \psi_2)(P, Z) \cdot |Z - \pi(P)|^2 = 0$ for all (P, Z); off $\mathrm{graph}(\pi)$ this gives $\psi_1 = \psi_2$, and by continuity on $\mathrm{graph}(\pi)$ as well. Since $C^1(\mathcal{P} \times \mathcal{Z}, \mathbb{R})$ is infinite-dimensional, so is $F(\tilde{f}, \pi)$. □

3.2 Joint Non-Identification

Theorem 3.2 (Joint non-identification)**.** *For any $\tilde{f} \in C^1(\mathcal{P}, \mathcal{P})$, the fiber $E(\tilde{f})$ contains infinitely many pairs (f, π) with* distinct *π-components. Specifically, for every $\pi_1, \pi_2 \in \Pi$ there exist $f_1, f_2 \in \mathcal{F}$ such that $(f_1, \pi_1) \in E(\tilde{f})$ and $(f_2, \pi_2) \in E(\tilde{f})$.*

Proof. Let $(f_1, \pi_1) \in E(\tilde{f})$ be any solution. Fix an arbitrary $\pi_2 \in \Pi$ and define

$$f_2(P, Z) \;=\; f_1\big(P,\ \pi_1(P) + (Z - \pi_2(P))\big). \tag{6}$$

Evaluating at $Z = \pi_2(P)$:

$$f_2(P, \pi_2(P)) = f_1\big(P, \pi_1(P) + \underbrace{(\pi_2(P) - \pi_2(P))}_{=0}\big) = f_1(P, \pi_1(P)) = \tilde{f}(P).$$

Since $\pi_1, \pi_2 \in C^1$ and $f_1 \in C^1$, the chain rule gives $f_2 \in \mathcal{F}$. As π_2 was arbitrary, the set of π-components in $E(\tilde{f})$ equals all of Π. □

3.3 Infinite Dimensionality of the Identification Fiber

Theorem 3.3 (Manifold structure of $E(\tilde{f})$)**.** *$E(\tilde{f})$ is an infinite-dimensional affine submanifold of $\mathcal{F}\times\Pi$. Its tangent space at any $(f,\pi)\in E(\tilde{f})$ contains a linear subspace isomorphic to $C^1(\mathcal{P}\times\mathcal{Z},\mathbb{R})\times C^1(\mathcal{P},\mathcal{Z})$.*

Proof. Define $\Phi\colon \mathcal{F}\times\Pi\to C^1(\mathcal{P},\mathcal{P})$ by $\Phi(f,\pi)=\mathcal{C}(f,\pi)$, so $E(\tilde{f})=\Phi^{-1}(\tilde{f})$. By Lemma 2.4, the Fréchet derivative at (f,π) is

$$D\Phi(f,\pi)[\delta f,\delta\pi](P)=\delta f(P,\pi(P))+\partial_Z f(P,\pi(P))\cdot\delta\pi(P).$$

Any δf that vanishes on graph(π) lies in $\ker D\Phi(f,\pi)$. By the construction in Theorem 3.1, the subspace of such δf is isomorphic to $C^1(\mathcal{P}\times\mathcal{Z},\mathbb{R})$, which is infinite-dimensional. The implicit function theorem on Banach spaces [6] then implies $E(\tilde{f})$ is a C^1 submanifold with the stated tangent structure. □

4 Partial Identification

While point identification is impossible, knowledge of $\tilde{f}$ does impose non-trivial restrictions on (f,π).

4.1 What Is Identified

Proposition 4.1 (Identified features)**.** *Let $(f,\pi)\in E(\tilde{f})$. The following quantities are identified from $\tilde{f}$ alone:*

(i) ***On-graph values:*** *$f(P,\pi(P))=\tilde{f}(P)$ for all $P\in\mathcal{P}$.*

(ii) ***Total derivative:*** *$D\tilde{f}(P)=\partial_P f(P,\pi(P))+\partial_Z f(P,\pi(P))\cdot D\pi(P)$ for all P.*

(iii) ***Fixed-point set:*** *$\{P^*\in\mathcal{P}:\tilde{f}(P^*)=P^*\}$ is identified as the set of steady states.*

The structural and policy components f, π individually are not identified.

4.2 Restoring Point Identification

Point identification is restored under any of the following conditions:

(A) Exogenous variation (interventions). If Z_t is set exogenously for a positive-measure subset of observations, the interventional distribution $P(P_{t+1}\mid \mathrm{do}(Z_t=z),P_t=P)=f(P,z)$ identifies f at the observed (P,z) pairs. Sufficient variation in both P and Z identifies f everywhere by standard nonparametric arguments [11].

(B) Parametric restrictions. If $f\in\{f_\theta:\theta\in\Theta\}$ with $\dim(\Theta)<\infty$, local identification may be restored via rank conditions on the Jacobian of Φ (Rothenberg [14]).

(C) Cross-equation restrictions. In multi-equation systems, symmetry or homogeneity restrictions (e.g., Slutsky symmetry) can reduce the identified set to a point or a lower-dimensional manifold.

5 Statistical Implications

5.1 Consistent Estimation of the Closed-Loop Operator

Suppose we observe an i.i.d. sample $\{(P_t,P_{t+1})\}_{t=1}^n$ from the stationary distribution induced by $\tilde{f}$. The closed-loop map is identified as the conditional mean regression:

$$\tilde{f}(P)\;=\;\mathbb{E}[P_{t+1}\mid P_t=P]. \tag{7}$$

Under Assumption 2.1 and the condition that $\tilde{f}\in W^{s,\infty}(\mathcal{P},\mathcal{P})$ (Sobolev class of smoothness s), standard series estimators achieve the minimax-optimal rate [16, 11]:

$$\left\|\hat{\tilde{f}}-\tilde{f}\right\|_\infty\;=\;O_p\Big(n^{-s/(2s+d)}\Big). \tag{8}$$

5.2 Impossibility of Consistent Structural Estimation

Theorem 5.1 (Statistical non-identification)**.** *Let $\hat{\tilde{f}}$ be any consistent estimator of $\tilde{f}$ based on $\{(P_t, P_{t+1})\}_{t=1}^n$. Then for any estimators $(\hat{f}, \hat{\pi})$ of the structural pair, for any $n \in \mathbb{N}$ and $\varepsilon > 0$, there exists $(f', \pi') \in E(\tilde{f})$ with*

$$\left\| f' - \hat{f} \right\|_{\mathcal{F}} > \varepsilon \quad \text{and} \quad \|\pi' - \hat{\pi}\|_{\Pi} > \varepsilon.$$

Hence no consistent estimator of (f, π) based on equilibrium data alone exists.

Proof. By Theorem 3.1, for any $(\hat{f}, \hat{\pi})$ and $\varepsilon > 0$, choose $\psi \in C^1(\mathcal{P} \times \mathcal{Z}, \mathbb{R})$ supported away from graph$(\hat{\pi})$ with $\|\psi\|_\infty$ sufficiently large. Set $f' = f_\psi$ as in (5) and $\pi' = \hat{\pi}$. Then $(f', \pi') \in E(\tilde{f})$ and $\left\| f' - \hat{f} \right\|_\infty \geq \left\| \psi \cdot |Z - \hat{\pi}|^2 \right\|_\infty > \varepsilon$ on the region where $\psi \neq 0$. Since such ψ exist for all $\varepsilon > 0$, the estimator $\hat{f}$ cannot be consistent. □

Remark 5.2. The result implies that confidence sets for (f, π) based on $\{(P_t, P_{t+1})\}$ alone have zero frequentist coverage. Structural estimation in feedback systems *requires* additional identifying variation beyond equilibrium paths.

6 Machine Learning and Algorithmic Identification

6.1 What Neural Networks Learn

Under standard universal approximation results [5, 1], sufficiently deep neural networks $g_\theta \colon \mathcal{P} \to \mathcal{P}$ trained to minimise the squared prediction loss

$$\hat{\theta} \;=\; \arg\min_\theta \sum_{t=1}^n \|P_{t+1} - g_\theta(P_t)\|^2$$

converge in probability to $\tilde{f}$ as $n \to \infty$. By Theorem 5.1, however, this convergence to $\tilde{f}$ does not imply convergence to f or π. Neural networks trained on equilibrium data learn the closed-loop operator, *not* the structural function.

6.2 Algorithmic Impossibility

Theorem 6.1 (Algorithmic non-identification)**.** *No deterministic algorithm $\mathcal{A} \colon C^1(\mathcal{P}, \mathcal{P}) \to \mathcal{F} \times \Pi$ can satisfy $\mathcal{A}(\tilde{f}) = (f^*, \pi^*)$ for every true data-generating pair (f^*, π^*). The map $\tilde{f} \mapsto (f, \pi)$ is not a well-defined function on $C^1(\mathcal{P}, \mathcal{P}) \to \mathcal{F} \times \Pi$.*

Proof. By Theorem 3.2, for any output $\mathcal{A}(\tilde{f}) = (f_0, \pi_0)$, there exists $(f_1, \pi_1) \in E(\tilde{f})$ with $\pi_1 \neq \pi_0$. Both pairs generate the same $\tilde{f}$. Any algorithm outputs at most one pair, so it fails to recover the true pair with probability at least $\frac{1}{2}$ over a uniform draw from $\{(f_0, \pi_0), (f_1, \pi_1)\}$. Since the argument holds for every algorithm, no deterministic algorithm achieves correct recovery. □

6.3 Non-identification Algorithm and Constructive Equivalence Class

Algorithm 1 provides a constructive procedure for generating any element of $E(\tilde{f})$ given a base solution and a perturbation function.

6.4 Causal Identification and Interventional Data

Within the do-calculus framework of Pearl [12] and Peters et al. [13], structural identification requires the *interventional* distribution:

$$P(P_{t+1} \mid \mathrm{do}(Z_t = z),\; P_t = P) \;=\; f(P, z), \tag{9}$$

which is directly readable from post-intervention data. The impossibility results are thus precisely delimited: they apply to the purely observational regime; exogenous interventions on Z resolve the non-identification.

Algorithm 1 Constructive Non-Identification: Generating $E(\tilde{f})$

Require: Closed-loop operator $\tilde{f} \in C^1(\mathcal{P}, \mathcal{P})$; base policy $\pi_0 \in \Pi$; perturbation $\psi \in C^1(\mathcal{P} \times \mathcal{Z}, \mathbb{R})$
Ensure: A pair $(f_\psi, \pi_0) \in E(\tilde{f})$ distinct from (f_0, π_0)

1: Compute the base structural function: $f_0(P, Z) \leftarrow \tilde{f}(P)$ ▷ trivially satisfies composition
2: Construct perturbation using the squared-distance kernel:

$$f_\psi(P, Z) \;\leftarrow\; \tilde{f}(P) + \psi(P, Z) \cdot |Z - \pi_0(P)|^2$$

3: **Verify:** evaluate $f_\psi(P, \pi_0(P))$ for a grid of $P \in \mathcal{P}$; confirm $\approx \tilde{f}(P)$ within tolerance δ
4: **Joint shift (optional):** for an alternative policy $\pi_1 \in \Pi$, set

$$f_1(P, Z) \leftarrow f_\psi\big(P,\ \pi_0(P) + (Z - \pi_1(P))\big)$$

5: **return** (f_ψ, π_0) **or** (f_1, π_1)

7 Numerical Illustration

7.1 Parametric Setup

Consider a scalar system ($d = k = 1$) with $\mathcal{P} = [0, 2]$, $\mathcal{Z} = [0, 1]$. The true data-generating pair is:

$$f^*(P, Z) = \alpha P + \beta Z + \gamma P Z, \qquad \pi^*(P) = \frac{\delta P}{1 + P}, \tag{10}$$

with $(\alpha, \beta, \gamma, \delta) = (0.70,\ 0.50,\ -0.10,\ 0.60)$. The resulting closed-loop map is:

$$\tilde{f}(P) = 0.70P + \frac{0.30P}{1 + P} - \frac{0.06P^2}{1 + P}. \tag{11}$$

Table 1 presents simulated observations from this system.

Table 1: Simulated observations from the closed-loop system. All rows are consistent with $\tilde{f}$ but do not identify (f^*, π^*).

P_t	P_{t+1} (obs.)	$\tilde{f}(P_t)$	$\pi^*(P_t)$
0.20	0.2817	0.2817	0.1000
0.40	0.5114	0.5114	0.1714
0.60	0.6968	0.6968	0.2250
0.80	0.8521	0.8521	0.2667
1.00	1.0000	1.0000	0.3000
1.20	1.1455	1.1455	0.3273
1.50	1.3688	1.3688	0.3600
2.00	1.7667	1.7667	0.4000

7.2 Observationally Equivalent Alternatives

For each alternative policy $\tilde{\pi}$, we construct $\tilde{f} \in \mathcal{F}$ such that $(\tilde{f}, \tilde{\pi}) \in E(\tilde{f})$:

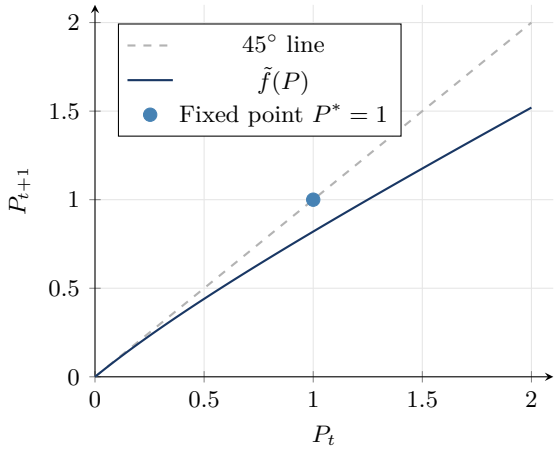
(1) Linear rule: $\tilde{\pi}_1(P) = 0.30P$. Set $\tilde{f}_1(P, Z) = f^*(P, \pi^*(P) + (Z - 0.30P))$. Then $\tilde{f}_1(P, \tilde{\pi}_1(P)) = f^*(P, \pi^*(P)) = \tilde{f}(P)$.

(2) Constant rule: $\tilde{\pi}_2(P) = 0.25$. Applying (6) gives $\tilde{f}_2$ that is distinct from f^* everywhere except at $P = P^*$ but generates the same equilibrium path.

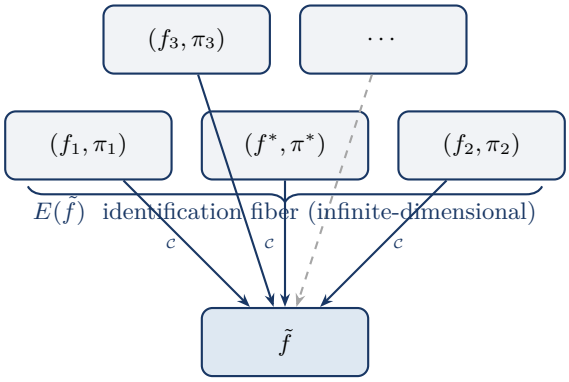
Both alternatives are *empirically indistinguishable* from the true model on equilibrium data, confirming the infinite multiplicity in Theorems 3.1–3.3.

7.3 Graphical Analysis

Figure 1a shows the closed-loop map $\tilde{f}$ and the 45-degree line. Figure 1b illustrates the identification fiber: the highlighted region represents the continuum of (f, π) pairs consistent with the same $\tilde{f}$.



(a) Closed-loop map $\tilde{f}$ and the 45-degree line.



(b) Schematic of the identification fiber $E(\tilde{f})$. All pairs map to the same closed-loop operator via $\mathcal{C}$.

Figure 1: Left: the closed-loop map for the numerical example, with fixed point at $P^* = 1$. Right: the infinite family of structural pairs (f, π) that are observationally identical.

7.4 Comparative Statics of the Closed-Loop Map

Figure 2 overlays the equilibrium trajectories generated by (f^*, π^*), $(\tilde{f}_1, \tilde{\pi}_1)$, and $(\tilde{f}_2, \tilde{\pi}_2)$. All three are identical, illustrating empirical indistinguishability.

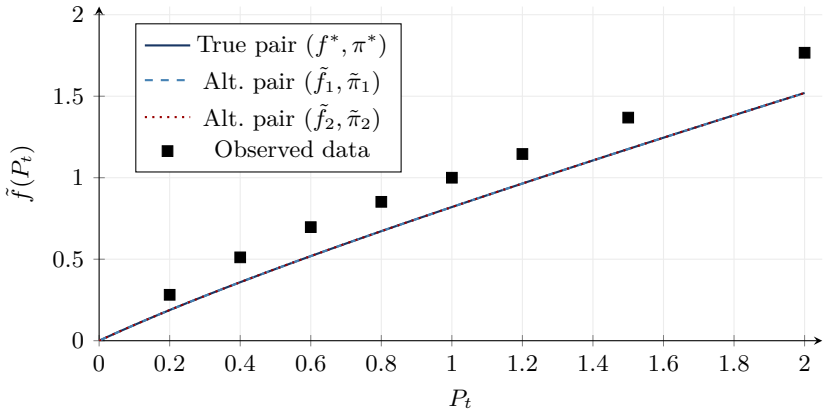


Figure 2: All three structural pairs generate the same equilibrium trajectory. The three curves are exactly coincident, illustrating Theorems 3.1–3.2.

8 Extensions

8.1 Stochastic Systems

Suppose the system is perturbed by independent shocks:

$$P_{t+1} = f(P_t, Z_t) + \varepsilon_t, \qquad Z_t = \pi(P_t) + \eta_t, \tag{12}$$

where (ε_t, η_t) are mean-zero with finite variance. The conditional mean $\mathbb{E}[P_{t+1} \mid P_t] = \tilde{f}(P_t)$ still identifies only the closed-loop operator; higher moments identify aspects of the shock distributions but not the structural decomposition. The impossibility theorems apply without modification to the conditional mean regression.

8.2 Infinite-Dimensional State Spaces

The results extend to Banach-valued state spaces (e.g., $\mathcal{P}$ a space of probability measures, relevant for heterogeneous-agent models). The composition operator $\mathcal{C}$ remains well-defined on appropriate function spaces, and the kernel-dimension argument in Theorem 3.3 carries through via the implicit function theorem on Banach manifolds [6].

8.3 Multi-Agent and Game-Theoretic Settings

In I-player games, each agent i implements a feedback rule π_i. The joint equilibrium map is $\tilde{f}(P) = f(P, \pi_1(P), \ldots, \pi_I(P))$. Our results imply that individual payoff functions—the structural primitives—are not identified from equilibrium play alone, connecting to the identification of games literature [2, 4].

8.4 Continuous-Time Systems

For systems governed by ODEs $\dot{P} = f(P, \pi(P))$, the right-hand side composite $\tilde{f}_{\text{ode}}(P) = f(P, \pi(P))$ is the identified object. Extension to SDEs follows by Itô's formula; the impossibility arguments are unchanged.

9 Conclusion

We have established that global identification of structural economic models is impossible under feedback control in a precise, comprehensive sense. The identification fiber $E(\tilde{f})$ is infinite-dimensional (Theorem 3.3); no consistent estimator of (f, π) exists on equilibrium data alone (Theorem 5.1); and no deterministic algorithm can recover the true structural pair from the closed-loop operator (Theorem 6.1).

These results unify and sharpen several strands of the literature. They provide a functional-analytic foundation for the Lucas Critique, connect it to the partial identification literature, and delineate exactly when and how identification can be restored—namely, through exogenous variation in the policy instrument, parametric restrictions, or cross-regime data.

For applied economists, the practical message is clear: structural estimation in feedback-stabilised systems requires either randomised interventions or theory-motivated parametric restrictions. Purely data-driven or machine-learning approaches cannot circumvent the impossibility established here.

Three directions for future work suggest themselves: *(i)* characterising partial identification bounds under shape restrictions (monotonicity, concavity) on f; *(ii)* deriving minimax lower bounds for the estimation of identified features such as the fixed-point set; and *(iii)* extending the framework to mean-field game equilibria arising in heterogeneous-agent macroeconomics.

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Glossary of Key Terms

Algorithmic non-identification
The property (Theorem 6.1) that no deterministic algorithm mapping the closed-loop operator $\tilde{f}$ to a structural pair (f, π) can guarantee recovery of the true data-generating pair.

Closed-loop operator
The composite map $\tilde{f} = \mathcal{C}(f, \pi) \colon \mathcal{P} \to \mathcal{P}$ defined by $\tilde{f}(P) = f(P, \pi(P))$. This is the only object directly observable from equilibrium data $\{(P_t, P_{t+1})\}$.

Composition operator
The operator $\mathcal{C} \colon \mathcal{F} \times \Pi \to C^1(\mathcal{P}, \mathcal{P})$ mapping structural pairs (f, π) to their closed-loop operator. Fréchet differentiable on the Banach space product $\mathcal{F} \times \Pi$ (Lemma 2.4).

Feedback policy
A mapping $\pi \colon \mathcal{P} \to \mathcal{Z}$ determining the policy instrument as a function of the current state. The feedback structure is the source of the identification impossibility.

Fréchet derivative
The infinite-dimensional generalisation of the derivative, applicable to operators between Banach spaces. Used to characterise the tangent space of $E(\tilde{f})$ in Theorem 3.3.

Global identification
The property that the structural pair (f, π) is the unique element of $\mathcal{F} \times \Pi$ consistent with the closed-loop operator $\tilde{f}$. Distinguished from *local* identification, which requires only local uniqueness near the true parameter.

Identification fiber $E(\tilde{f})$
The set $\{(f, \pi) \in \mathcal{F} \times \Pi : f(P, \pi(P)) = \tilde{f}(P) \, \forall P\}$ of all structural pairs observationally identical to the true data-generating pair. An infinite-dimensional affine submanifold of $\mathcal{F} \times \Pi$ (Theorem 3.3).

Interventional distribution
The distribution of P_{t+1} under the intervention $\mathrm{do}(Z_t = z)$, denoted $P(P_{t+1} \mid \mathrm{do}(Z_t = z), P_t = P) = f(P, z)$. Availability of interventional data restores point identification of f.

Lucas Critique

The argument of Lucas [8] that reduced-form relationships depend on the policy rule π and are therefore non-invariant to policy changes. Our results provide a formal functional-analytic proof of this critique's impossibility character.

Partial identification

The situation in which the data restrict (f, π) to a set larger than a singleton. Formalised here by characterising the identified set as $E(\tilde{f})$ under equilibrium data; see Manski [9, 10].

Structural transition operator

The function $f\colon \mathcal{P} \times \mathcal{Z} \to \mathcal{P}$ governing how the state evolves given the current state and policy instrument. The primitive object of interest; non-identifiable from equilibrium data alone.

The End